

Project Design Phase-II

Technology Stack (Architecture & Stack)

- **Date:** 18 June 2026
- **Project Name:** Shopez

Technical Architecture

Shopez uses a MERN (MongoDB, Express, React, Node.js) 3-tier architecture:

- **Presentation Layer:** React.js with CSS styling.
- **Application Layer:** Express.js server on Node.js environment.
- **Database Layer:** MongoDB (Cloud Atlas) with a JSON fallback.

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Web application front-end	HTML5, CSS3, JavaScript, React.js (Vite)
2	Application Logic	RESTful API server handling authentication, products, orders, and stats	Node.js, Express.js
3	Database	Core persistent store for products, users, and orders	MongoDB (Mongoose)
4	Fallback Storage	Fallback database if MongoDB is unreachable	Local JSON files (products.json, orders.json)
5	Authentication	Token-based security and authorization	JSON Web Tokens (JWT), bcryptjs
6	Deployment Server	Hosting environment	Local Node.js server / Render Cloud Hosting

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	React, Express, and Mongoose form the backbone	React.js v18+, Express v4+
2	Security Implementations	Password hashing, secure routes, login authorization controls	bcryptjs, JWT
3	Scalable Architecture	3-tier separation enables independent frontend & backend scaling	REST Architecture

S.No	Characteristics	Description	Technology
4	Availability	Local database fallback ensures system runs offline or on database loss	JSON file persistence
5	Performance	Clean API routing and static assets delivery from Vite dev server	Vite, Node Cluster